ABB Robotics IRB4600 Product Manual - System Specifications

Model: IRB4600-40/2.55 Manipulator with IRC5 Controller.

Payload: 40 kg rated, maximum reaching distance: 2.55 meters.

Number of Axes: 6 articulated rotary axes with AC brushless servomotors.

Protection Rating: Standard IP67 casting enclosure suited for harsh manufacturing environments.

Fieldbus Interface: Dual-port PROFINET IO Device connecting to Cell Master PLC01.

Operating Temperature Range: 5 deg C to 45 deg C continuous industrial duty cycle.

ABB Robotics IRB4600 - Axis Calibration and Homing Procedures

Calibration offsets must be synchronized with the IRC5 controller resolver memory.

If battery voltage in the Serial Measurement Board (SMB) drops below 3.4V, revolution counters are lost.

Refer to fault manual FLT-001 Robot Home Position for recalibration steps.

Align physical vernier zero marks on Axis 1 through Axis 6 before updating revolution counters.

Mechanical stops are fitted on Axis 1, 2, and 3 with polyurethane buffer dampeners.

Home position confirmation limit switch must actuate reliably within +/- 1.0 mm tolerance.

ABB Robotics IRB4600 - Wrist Tooling and Gripper Interfacing

End-of-Arm Tooling connects through the Axis 6 mounting flange (ISO 9409-1-100-6-M8).

Pneumatic gripper clamping uses 5/2 directional solenoid valve island on SMC manifold.

Clamp confirmation signals are monitored via magnetic reed switches wired to digital inputs I:0/15 and I1.15.

If confirmation is not received within 1.2 seconds, a gripper timeout fault occurs (see Faults/Robot Gripper.md).

Use high-flex robotic polyurethane cabling for dress pack to prevent internal conductor breakage.

ABB Robotics IRB4600 - Maintenance Protocols and Inspection Intervals

500-Hour Inspection: Inspect dress pack cabling along Axis 5 and 6 for torsional fatigue.

6000-Hour Overhaul: Drain and replenish synthetic gearbox grease on Axis 1, 2, and 3 gearsets.

Replace SMB lithium battery pack (part number 3HAC044075-001) every 24 months.

Check brake holding torque annually using RobotStudio automatic BrakeCheck routine.

Adhere to lockout tagout procedures in Procedures/Lockout Tagout Cell A.md before entering safety envelope.